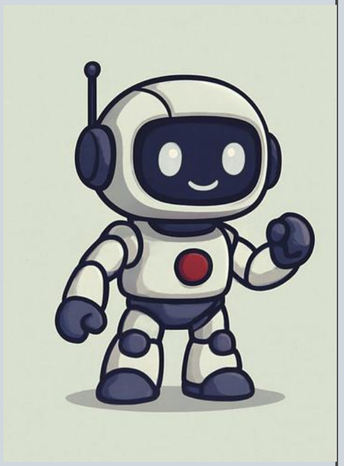
SYNAPSE -IIITM

Presented By : Synapse Team



Today's Agenda

01 What is AI?

02 Why are humans better than AI?

03 Why AI is better than humans?

04 What is the career path of AI?

05 Real-world example

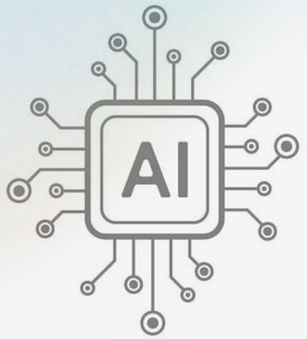
06 Setup : Python , VS Code , Jupyter Notebook , Anaconda

What is Artificial Intelligence?

AI is the ability of a computer system to perform tasks that normally require human intelligence.

Human intelligence includes:

- Learning
- Understanding
- Recognizing patterns
- Making decisions
- Solving problems
- Understanding language



What is Artificial Intelligence?

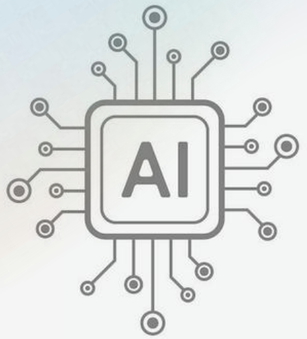
AI can also act like human.

Example: patterns

World Robot Conference







AI You Already Use (Today)

You Interact with AI constantly :

- Face Unlock on your phone – recognizes your face instantly
- Instagram Reels & YouTube recommendations – learns what you like
- Google Search & Autocomplete – predicts what you'll search
- Music app playlist – suggests songs based on your taste

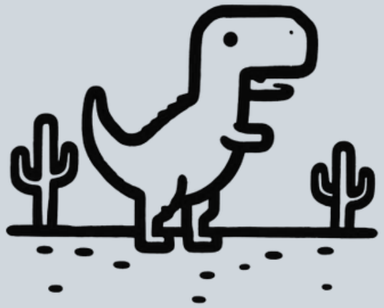


Why Humans Are Better than AI?

If AI is so powerful, why don't we replace humans completely?

- **Creativity**
- **Common sense**
- **Emotions**
- **Adaptability**
- **Goals and values**

If your friend is sad, ChatGPT can generate a message to comfort them. But does ChatGPT actually experience friendship or sadness?



Why AI is Better than Humans?

Can you check my entire files in one hour?

- **Speed**
- **Repetition**
- **Large-scale data processing**
- **Pattern detection**
- **24/7 operation**

AI is not always better than humans. AI + Humans can be better than either alone.

What Is The Career Path In AI?

Core Technical Roles 🔧

- **Machine Learning Engineer — Algorithms, TensorFlow/PyTorch**
- **AI Research Scientist — Experiments, innovation, theory**
- **Computer Vision Engineer — Image processing, autonomous systems**
- **Robotics Engineer — Physical AI systems, embedded**

Implementation & Systems 💻

- **Data Scientist — Data analysis, insights**
- **AI Software Engineer — Production deployment, scalability**
- **AI Hardware Engineer — GPUs, TPUs optimization**
- **NLP Engineer — Language models, chatbots**

Business & Leadership 📊

- **AI Product Manager — Strategy, product lifecycle**
- **Business Intelligence Developer — Dashboards, analytics**
- **AI Ethics Specialist — Bias, fairness, responsibility**

Roles In AI

01 ML Engineer

Builds and optimizes ML models for production—writes training code, handles data pipelines, deploys models to real systems, ensures performance and reliability.

02 ML Ops Engineer

Manages the infrastructure and lifecycle of ML systems in production—monitors models, automates retraining, handles versioning, scaling, and CI/CD for ML pipelines.

03 AI Engineer

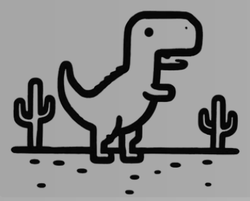
Broader role focusing on integrating AI systems into products—works with LLMs, multimodal models, prompt engineering, RAG systems, and building full AI applications (not just training models).

04 Data Scientist

Analyzes data to answer business questions and generate insights—does exploratory data analysis, statistical modeling, A/B testing, and sometimes builds prototype models before engineers productionize them.

05 AI Research Scientist

Advances the frontier of AI/ML—develops new algorithms, publishes papers, conducts experiments, and creates foundational techniques that eventually get used by engineers and other roles.



AI Teachable Machines

1.Gather DATA

Feed examples (images/sounds/poses)

2.TRAIN

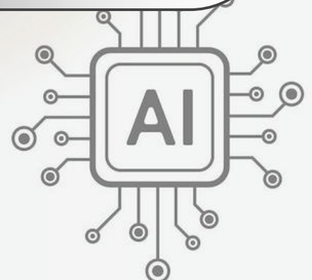
Synapses strengthen via pattern recognition

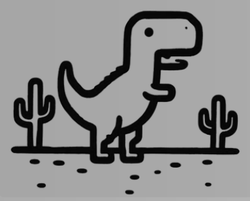
3.TEST

Vaidate accuracy on new data

4.DEPLOY

Export to apps , robots , microcontrollers





QuickDraw

1.Random item

Feed examples (images)

2.Draw

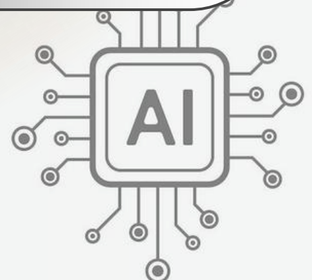
draw image of given object

3.Predict

AI predict your drawing by given item

4.Train

train model for predict more accurately in future



Human or Not

Real World Examples

Project Euphonia : Facial Gestures

A model that can communicate using facial gestures to trigger sounds.



Teachable Snake : Web Cam & Paper

A model that can turn your web cam and a piece of paper into a game controller.



Teachable Sorter : Custom ML models

A model that you can teach to rapidly recognize and sort objects using your own custom machine learning models.



Install Python

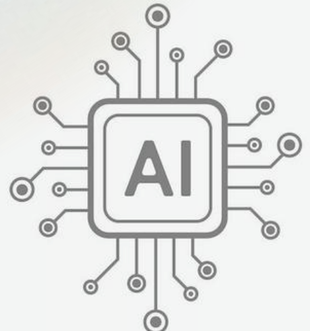
1. **Visit: Python.org**
2. **For Mac:**
    a. Download package
    b. follow step-by-step prompts in setup window
3. **For Window :**
    a. Download .exe file
    b. click the box : Add python.exe to PATH
    c. Install Now
4. **CMD : python --version/python3 --version**

1. INSTALL: VS CODE

2. INSTALL: JUPYTER

   NOTEBOOK/EXTENSION

3. INSTALL: ANCONDA







Thank you

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